

PRO2 (CCNA)

Enterprise Network Design & Implementation

Project 2 – August 2026

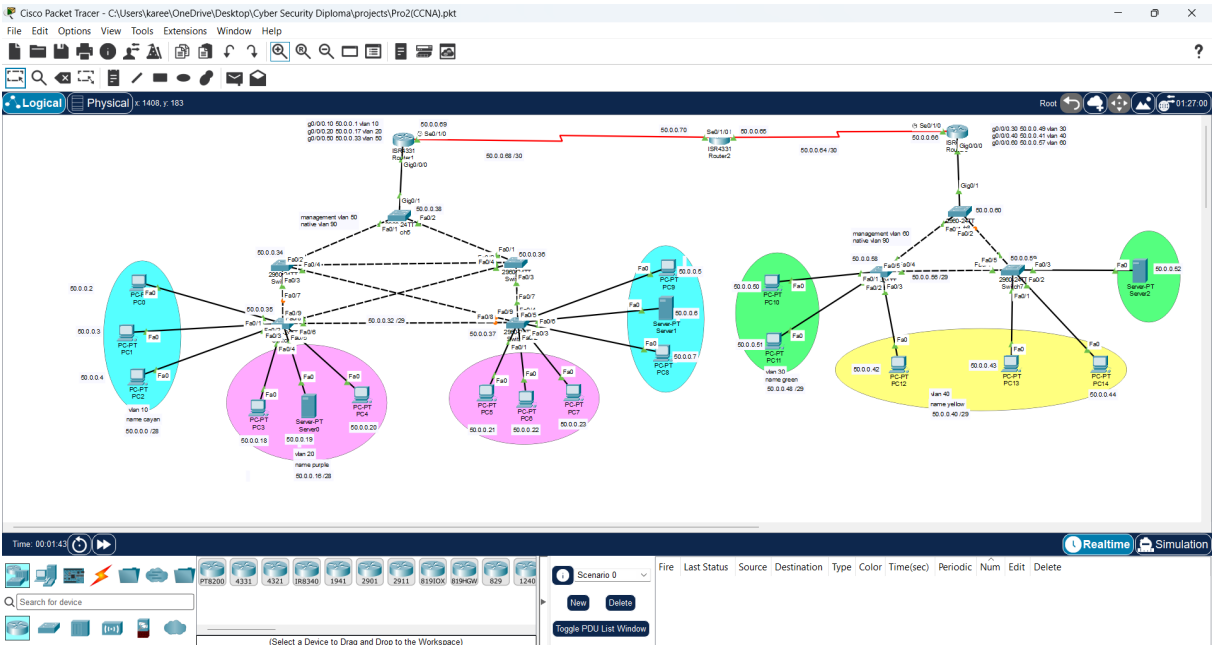


Figure 1 – Complete Logical Topology (Left Network • Core Routers • Right Network)

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Project	Pro2 (CCNA) – Multi-Site Network with OSPF & Inter-VLAN Routing
Technologies	VLSM • VLANs • Trunking • Inter-VLAN Routing • OSPF • SSH / Telnet • Device Security
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1. Project Overview

This document describes the design, configuration and verification of **Pro2 (CCNA)**, a multi-site enterprise network built in August 2026. The topology consists of a Left Network (Router1 with VLANs 10, 20 and management 50), a Right Network (Router3 with VLANs 30, 40 and management 60), and a Core path through Router2 using serial links. OSPF provides dynamic routing between the three routers. Native VLAN is 90 on all switches.

Site	Role	Key Features
Left Network	Router1 + SW1–SW5	VLANs 10/20/50, Inter-VLAN Routing, Mgmt VLAN 50
Right Network	Router3 + SW6–SW8	VLANs 30/40/60, Inter-VLAN Routing, Mgmt VLAN 60
Core	Router2 (transit)	Serial links /30, OSPF Area 0

2. Network Topology

The network is divided into two campus sides connected through three routers over serial links.

2.1 Left Network (Router1 side)

Router1 performs Inter-VLAN routing for VLAN 10 (cyan), VLAN 20 (purple) and management VLAN 50. Five switches (SW1–SW5) form the access/distribution layer. Native VLAN is 90. Management subnet is 50.0.0.32/29.

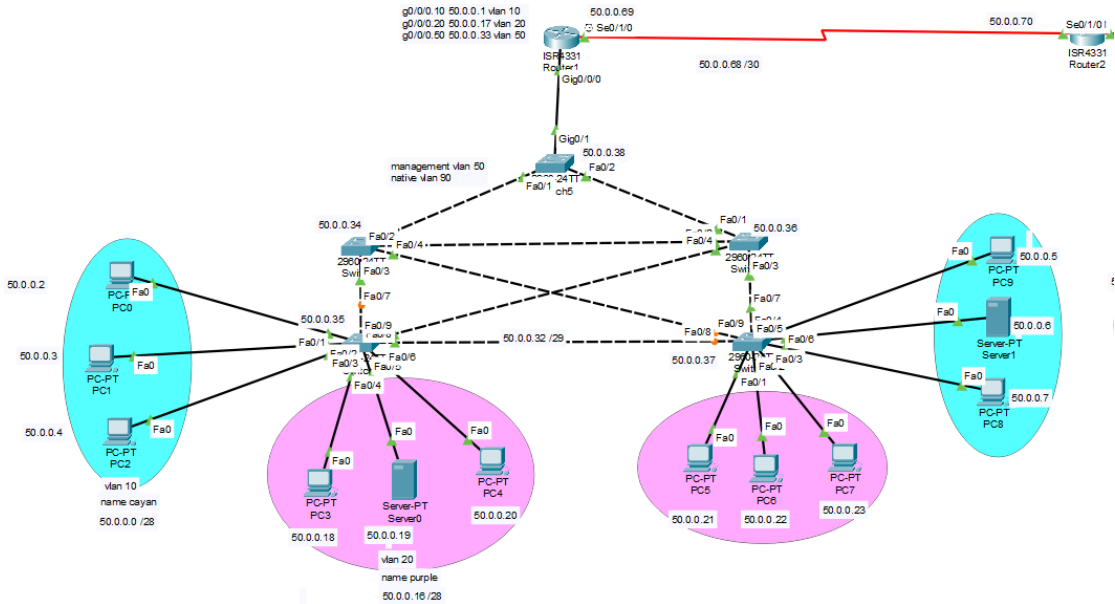


Figure 2 – Left Network detailed topology (VLANs 10/20/50, SW1–SW5, Router1)

2.2 Right Network (Router3 side)

Router3 performs Inter-VLAN routing for VLAN 30 (green), VLAN 40 (yellow) and management VLAN 60. Three switches (SW6–SW8) serve the access layer. Native VLAN is 90. Management subnet is 50.0.0.56/29.

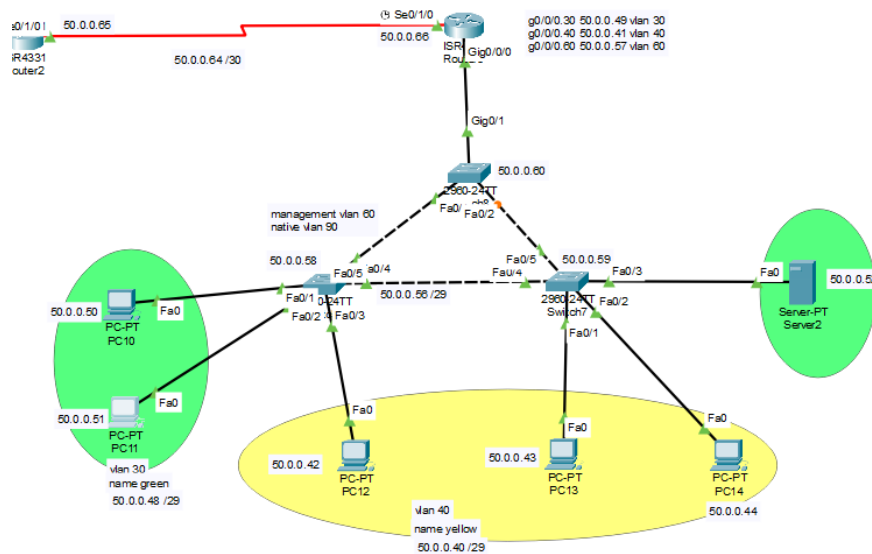


Figure 3 – Right Network detailed topology (VLANs 30/40/60, SW6–SW8, Router3)

2.3 Serial / Core Links

Router1 connects to Router2 via 50.0.0.68/30. Router2 connects to Router3 via 50.0.0.64/30. All three routers run OSPF and form FULL adjacencies.

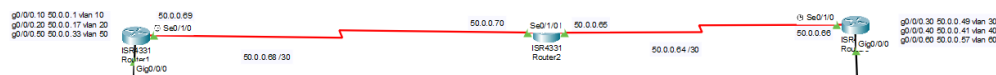


Figure 4 – Serial links between Router1 – Router2 – Router3

3. Requirements Checklist

The following requirements were implemented and verified:

#	Requirement	Status
1	Subnetting of 50.0.0.0/8	Done
2	IP assignment to all end devices	Done
3	VLANs on switches (same color = same VLAN)	Done
4	Native VLAN 90 on all switches	Done
5	Management VLAN 50 (Left) with dedicated subnet	Done
6	Management VLAN 60 (Right) with dedicated subnet	Done
7	Access ports on switches	Done
8	Trunk ports on switches	Done
9	Inter-VLAN Routing	Done
10	OSPF on Routers	Done
11	Console & Enable passwords	Done
12	Telnet & SSH on all devices	Done

4. Subnetting & IP Addressing Plan

Base: 50.0.0.0/8 was subdivided with VLSM to match each VLAN and serial link requirement.

Purpose	Network	Mask	Gateway
VLAN 10 (cyan) – Left	50.0.0.0/28	/28	50.0.0.1
VLAN 20 (purple) – Left	50.0.0.16/28	/28	50.0.0.17
Mgmt VLAN 50 – Left	50.0.0.32/29	/29	50.0.0.33
VLAN 40 (yellow) – Right	50.0.0.40/29	/29	50.0.0.41
VLAN 30 (green) – Right	50.0.0.48/29	/29	50.0.0.49
Mgmt VLAN 60 – Right	50.0.0.56/29	/29	50.0.0.57
Serial R2–R3	50.0.0.64/30	/30	—
Serial R1–R2	50.0.0.68/30	/30	—

5. Complete Device IP Addressing Tables

5.1 Left Network – End Devices

Device	IP	VLAN	Gateway
PC0	50.0.0.2	10 (cayan)	50.0.0.1
PC1	50.0.0.3	10 (cayan)	50.0.0.1
PC2	50.0.0.4	10 (cayan)	50.0.0.1
PC9	50.0.0.5	10 (cayan)	50.0.0.1
Server1	50.0.0.6	10 (cayan)	50.0.0.1
PC8	50.0.0.7	10 (cayan)	50.0.0.1
PC3	50.0.0.18	20 (purple)	50.0.0.17
Server0	50.0.0.19	20 (purple)	50.0.0.17
PC4	50.0.0.20	20 (purple)	50.0.0.17
PC5	50.0.0.21	20 (purple)	50.0.0.17
PC6	50.0.0.22	20 (purple)	50.0.0.17
PC7	50.0.0.23	20 (purple)	50.0.0.17

5.2 Left Network – Switches (Mgmt VLAN 50)

Switch	IP	Mask	Gateway
SW1	50.0.0.34	/29	50.0.0.33
SW2	50.0.0.35	/29	50.0.0.33
SW3	50.0.0.36	/29	50.0.0.33
SW4	50.0.0.37	/29	50.0.0.33
SW5	50.0.0.38	/29	50.0.0.33

5.3 Right Network – End Devices

Device	IP	VLAN	Gateway
PC12	50.0.0.42	40 (yellow)	50.0.0.41
PC13	50.0.0.43	40 (yellow)	50.0.0.41
PC14	50.0.0.44	40 (yellow)	50.0.0.41
PC10	50.0.0.50	30 (green)	50.0.0.49
PC11	50.0.0.51	30 (green)	50.0.0.49
Server2	50.0.0.52	30 (green)	50.0.0.49

5.4 Right Network – Switches (Mgmt VLAN 60)

Switch	IP	Mask	Gateway
SW6	50.0.0.58	/29	50.0.0.57
SW7	50.0.0.59	/29	50.0.0.57
SW8	50.0.0.60	/29	50.0.0.57

5.5 Routers

Router	Interface	IP Address	Notes
Router1	G0/0/0.10	50.0.0.1/28	VLAN 10 gateway
Router1	G0/0/0.20	50.0.0.17/28	VLAN 20 gateway
Router1	G0/0/0.50	50.0.0.33/29	Mgmt VLAN 50 gateway
Router1	S0/1/0	50.0.0.69/30	Link to Router2
Router2	S0/1/0	50.0.0.70/30	Link to Router1
Router2	S0/1/1	50.0.0.65/30	Link to Router3
Router3	G0/0/0.30	50.0.0.49/29	VLAN 30 gateway
Router3	G0/0/0.40	50.0.0.41/29	VLAN 40 gateway
Router3	G0/0/0.60	50.0.0.57/29	Mgmt VLAN 60 gateway
Router3	S0/1/0	50.0.0.66/30	Link to Router2

6. Implementation Steps

Step 1 – Subnetting & IP Assignment

The base network 50.0.0.0/8 was divided using VLSM. Each VLAN received an appropriately sized subnet. All end devices and switch management interfaces were assigned static IP addresses according to the plan above. Serial links received /30 networks.

Step 2 – VLANs, Access Ports & Trunks

VLANs were created and named on all switches: 10 (cyan), 20 (purple), 30 (green), 40 (yellow), 50 (management – Left), 60 (management – Right), 90 (native). Access ports were assigned to the correct VLANs. Trunk links were configured with native VLAN 90 and allowed VLANs 10,20,30,40,50,90 (and 60 on the right side).

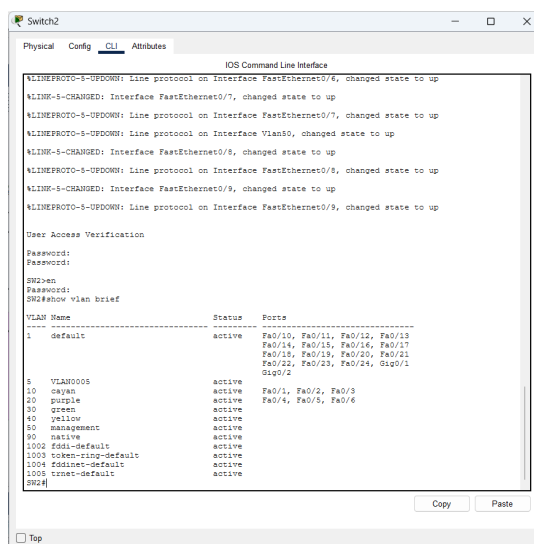


Figure 5 – show vlan brief on SW2.

What was done: VLANs 10,20,30,40,50,90 created and ports assigned.

Why it was used: VLANs segment the network by department/function and reduce broadcast domains.

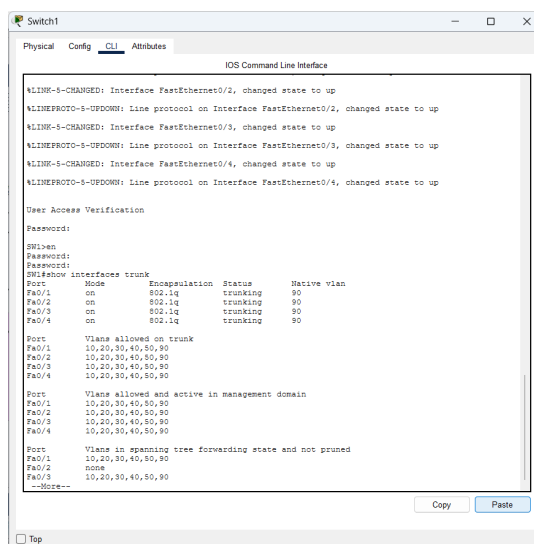


Figure 6 – show interfaces trunk on SW1.

What was done: Trunk ports configured with native VLAN 90; required VLANs allowed.

Why it was used: Trunks carry multiple VLANs between switches while native VLAN 90 improves security.

Step 3 – Inter-VLAN Routing

Router-on-a-stick was configured on Router1 (sub-interfaces for VLANs 10, 20, 50) and Router3 (sub-interfaces for VLANs 30, 40, 60). This enables communication between different VLANs on the same site.

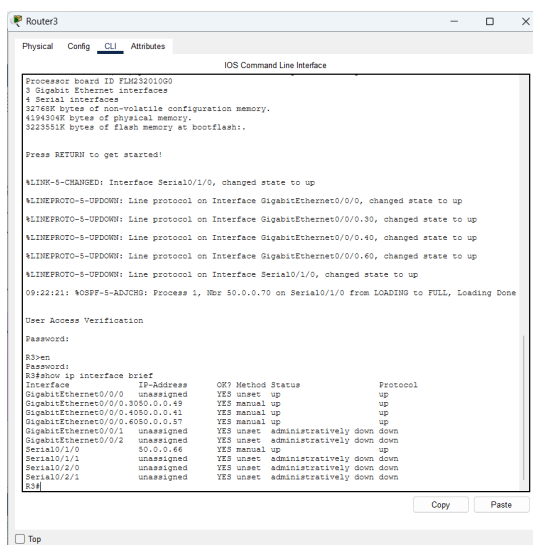


Figure 7 – show ip interface brief on Router3.

What was done: Sub-interfaces G0/0/0.30, .40, .60 configured with correct IPs; Serial0/1/0 up.

Why it was used: Inter-VLAN routing allows hosts in different VLANs to communicate through the router.

Step 4 – OSPF Routing

OSPF was enabled on Router1, Router2 and Router3. All routers formed FULL adjacencies over the serial links. Remote networks (Left ↔ Right) became reachable through OSPF-learned routes.

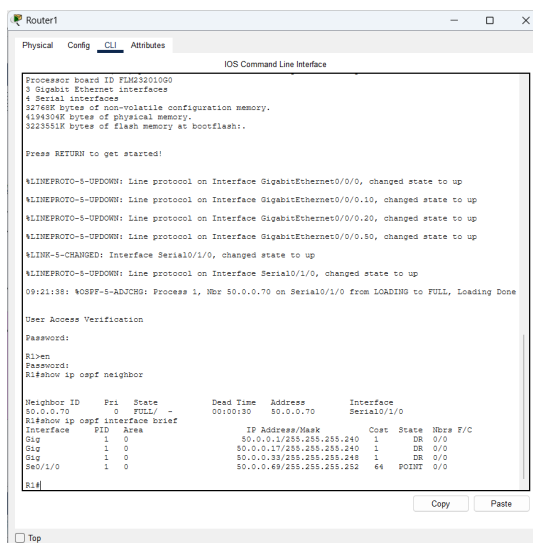
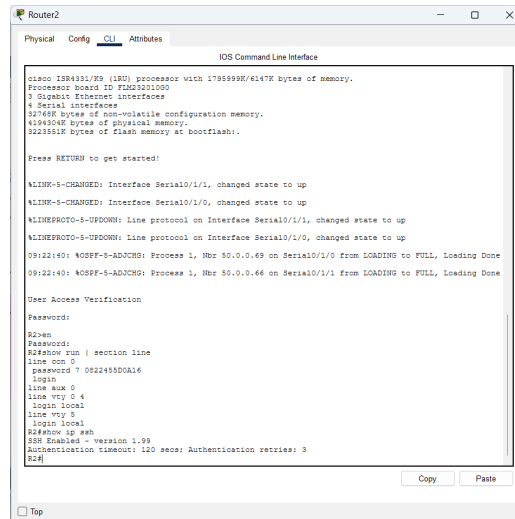


Figure 8 – show ip ospf neighbor & show ip ospf interface brief on Router1.

What was done: OSPF neighbor 50.0.0.70 in FULL state; interfaces in Area 0.

Why it was used: OSPF provides dynamic routing so the network adapts automatically to changes.



```
Router2
Physical Config CLI Attributes
IOS Command Line Interface

disco ISR4321/F9 (R9) processor with 1795999K/4147K bytes of memory.
Processor board ID F4C13P11000
3 Gigabit Ethernet interfaces
4 Serial interfaces
32768K bytes of non-volatile configuration memory.
4194504K bytes of physical memory.
3223551K bytes of flash memory at bootflash.

Press RETURN to get started!

%LINK-5-CHANGED: Interface Serial0/1/1, changed state to up
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up
09:22:40: %OSPF-5-ADJCHG: Process 1, Nbr 50.0.0.69 on Serial0/1/0 from LOADING to FULL, Loading Done
09:22:40: %OSPF-5-ADJCHG: Process 1, Nbr 50.0.0.66 on Serial0/1/1 from LOADING to FULL, Loading Done

User Access Verification

Password:
%>en
Password:
%#show run | section line
line con 0
password 7 05224550A16
login
line aux 0
line vty 0 4
login local
line vty 5
login local
%#show ip ssh
SSH Enabled - version 1.99
Authentication timeout: 120 secs; Authentication retries: 3
%#
```

Figure 9 – Router2: OSPF adjacencies FULL with both neighbors + SSH enabled.
What was done: OSPF FULL with 50.0.0.69 and 50.0.0.66; SSH version 1.99 active.
Why it was used: Confirms Core transit is operational and management access is secure.

Step 5 – Device Security (Passwords, Telnet & SSH)

Console, enable-secret and VTY passwords were configured on all routers and switches. SSH was enabled (RSA keys generated, domain name set, transport input ssh). Telnet was also available where required. Remote SSH login was verified from an end device.

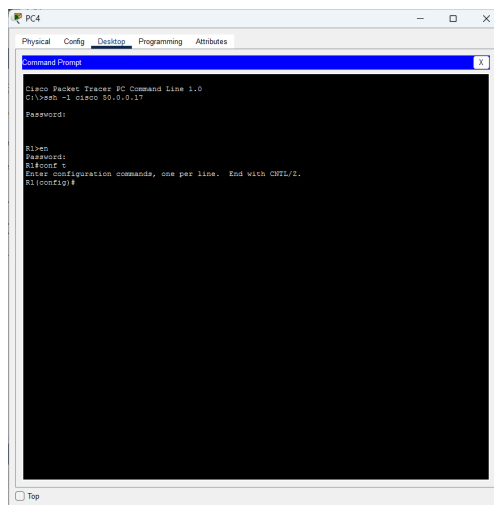


Figure 10 – Successful SSH login from PC4 to Router1 (50.0.0.17).

What was done: SSH session established; privileged and config modes reached.

Why it was used: SSH encrypts management traffic and is preferred over clear-text Telnet.

7. End-to-End Connectivity Verification

From PC0 (VLAN 10 – Left) the following connectivity tests were performed successfully with 0% packet loss:

- Same VLAN: ping 50.0.0.3 (PC1)
- Inter-VLAN (Left): ping 50.0.0.18 (PC3 in VLAN 20)
- Cross-site (Right): ping 50.0.0.50 (PC10 in VLAN 30)

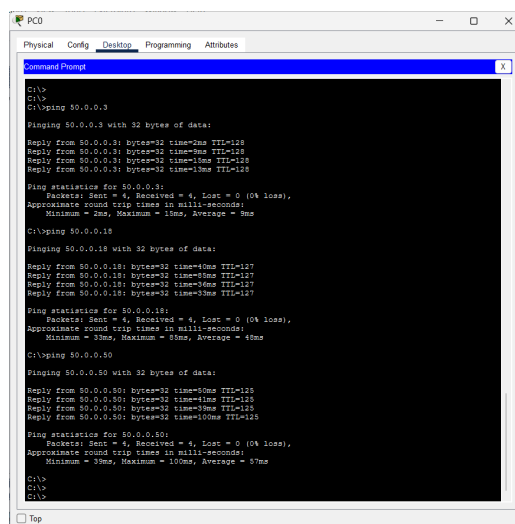


Figure 11 – Ping tests from PC0: same VLAN, inter-VLAN and cross-site – all successful (0% loss).

What was done: End-to-end connectivity verified across VLANs and across sites.

Why it was used: Confirms Inter-VLAN routing, OSPF and addressing are all working correctly.

8. Important Show Commands

The following commands were used for verification and are essential for troubleshooting:

8.1 Switches

```
show vlan brief
show interfaces trunk
show interfaces switchport
show ip interface brief
show running-config | include hostname
show running-config | section line
```

8.2 Routers – Interfaces & Addressing

```
show ip interface brief
show ip interface
show running-config interface GigabitEthernet0/0/0.10
```

8.3 Routers – OSPF

```
show ip ospf neighbor
show ip ospf interface brief
show ip route ospf
show ip protocols
show ip ospf database
```

8.4 Security & SSH

```
show ip ssh
show users
show running-config | section line
show running-config | include username|enable|secret
```

8.5 Connectivity Tests (from PCs)

```
ping <destination>
tracert <destination>
ssh -l cisco <router-ip>
```

9. Summary of Work Completed

1. **Subnetting** – 50.0.0.0/8 divided with VLSM into VLAN and serial subnets.
2. **IP Assignment** – Static IPs applied to all PCs, servers and switch management interfaces.
3. **VLANs & Trunking** – VLANs 10/20/30/40/50/60/90 created; access and trunk ports configured; native VLAN 90.
4. **Inter-VLAN Routing** – Router-on-a-stick on Router1 and Router3 for local VLAN communication.
5. **OSPF** – Area 0 on all three routers; FULL adjacency confirmed; cross-site reachability achieved.
6. **Security** – Console, enable and VTY passwords; SSH enabled and verified from end devices.

10. Conclusion

Pro2 (CCNA) successfully implements a dual-campus network with proper VLSM addressing, VLAN segmentation, Inter-VLAN routing, OSPF dynamic routing and secure device management. All verification commands and connectivity tests (same-VLAN, inter-VLAN and cross-site) confirm that the network is fully operational.

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